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Which one of the following statements is NOT CORRECT?

- ① A well-conditioned matrix has a condition number close to 1
- ② An ill-conditioned matrix has a large condition number
- ③ The inverse of a well-conditioned matrix can be computed accurately
- ④ A non-invertible matrix has a condition number close to 1

Theoretical Solution

The **condition number** of an invertible matrix $\mathbf{A}$, denoted as $\kappa(\mathbf{A})$, is defined as:

$$\kappa(\mathbf{A}) = \|\mathbf{A}\| \left\| \mathbf{A}^{-1} \right\| \quad (1)$$

where,

$$\|\mathbf{A}\| = \text{norm of matrix } \mathbf{A}$$

It measures how much the output of a linear system $\mathbf{Ax} = b$ can change for a small change in the input data.

Theoretical Solution

The most theoretically important and widely used condition number is the one defined with respect to the **matrix 2-norm** (also known as the spectral norm).

The 2-norm of a matrix $\mathbf{A}$, $\|\mathbf{A}\|_2$, is its largest singular value, $\sigma_{\max}(\mathbf{A})$.

The 2-norm of its inverse, $\|\mathbf{A}^{-1}\|_2$, is the reciprocal of the smallest singular value of $\mathbf{A}$, $1/\sigma_{\min}(\mathbf{A})$.

Therefore, the **spectral condition number**, denoted $\kappa_2(\mathbf{A})$, is defined as the ratio of the largest to the smallest singular value of the matrix:

$$\kappa_2(\mathbf{A}) = \frac{\sigma_{\max}(\mathbf{A})}{\sigma_{\min}(\mathbf{A})} \quad (2)$$

where:

- $\sigma_{\max}(\mathbf{A})$ is the largest singular value of $\mathbf{A}$.
- $\sigma_{\min}(\mathbf{A})$ is the smallest singular value of $\mathbf{A}$.

Theoretical Solution

For a symmetric positive definite matrix, the singular values are the same as the eigenvalues. In this special case, the definition simplifies to the ratio of the largest to the smallest eigenvalue:

$$\kappa_2(\mathbf{A}) = \frac{\lambda_{\max}(\mathbf{A})}{\lambda_{\min}(\mathbf{A})} \quad (\text{for symmetric positive definite } \mathbf{A}) \quad (3)$$

What the Condition Number Measures:

The condition number provides a measure of the sensitivity of the solution of a linear system of equations $\mathbf{Ax} = \mathbf{b}$ to errors or perturbations in the input data ($\mathbf{A}$ or $\mathbf{b}$).

Specifically, it bounds the maximum possible relative error in the solution $\mathbf{x}$ for a given relative error in the input vector $\mathbf{b}$. The fundamental inequality is:

$$\frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|} \leq \kappa(\mathbf{A}) \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|} \quad (4)$$

Interpretation:

- If $\kappa(\mathbf{A})$ is small (close to 1), the matrix is **well-conditioned**. This means that small relative errors in $\mathbf{b}$ will only lead to small relative errors in the solution $\mathbf{x}$. The solution is stable and reliable.
- If $\kappa(\mathbf{A})$ is large, the matrix is **ill-conditioned** or **poorly-conditioned**. This means that a tiny relative error in $\mathbf{b}$ can be magnified into a very large relative error in the solution $\mathbf{x}$. The solution is unstable and may be unreliable from a numerical perspective.

- 1) **A well-conditioned matrix has a condition number close to 1:** This is **CORRECT**. The smallest possible condition number is 1 (for example, the identity matrix). A value close to 1 indicates that the matrix is numerically stable.
- 2) **An ill-conditioned matrix has a large condition number:** This is **CORRECT**. This is the definition of an ill-conditioned matrix. A large condition number means that small errors in the input can lead to very large errors in the solution.

- 3) **The inverse of a well-conditioned matrix can be computed accurately:** This is **CORRECT**. Because a well-conditioned matrix is not sensitive to small perturbations, numerical algorithms used to compute its inverse will produce results with high accuracy.
- 4) **A non-invertible matrix has a condition number close to 1:** This is **NOT CORRECT**. A non-invertible (or singular) matrix does not have an inverse $\mathbf{A}^{-1}$. For the purposes of conditioning, its inverse is considered to have an infinite norm. Therefore, the condition number of a non-invertible matrix is defined to be **infinity** (∞). An infinite condition number is the largest possible value, representing the worst possible conditioning, not the best (which is close to 1).